Project Synopsis:

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PackCheck: Portable EV Battery Diagnostic and Charging Tool

Description

PackCheck is a handheld electronic diagnostic and cell-balancing tool targeted towards EV repair shops. It can connect directly to standard 48V–72V Lithium-ion battery packs with and have variety of interfacing options. It can identify individual cell voltages, identify weak cells, measure State of Health and internal resistance. The device has an option to balance the cells to equalize the power distribution as well as charge the battery pack. It also has a 2-inch LCD for showing diagnostics. The product serves as a "first-aid" kit for EV batteries, which help to extend the usefulness of the battery.

Techno-economic feasibility & Survey

Primary electronic components (microcontroller, sensing ICs, MOSFETs, etc.) are off-the-shelf and readily available from trusted distributors like DigiKey. PCB fabrication can be outsourced to fab houses (local, or bulk ordering from Chinese manufacturers). At present ordering from maufacturers like JLC PCB is most cost effective. They are also very reliable. Through this, internal processes can be cut to just final assembly, testing, and firmware flashing.

In terms of economic feasibility, we have kept a high contribution margin per unit, allowing the business to hit BEP by 6 days, with an estimated investment payback period of 12 months.

We identified similar products in the market, but they are in the price range of ₹9k-₹15k. Hence, our product has a attractive price option.

Cost & Profit margin

To reduce cost, we plan to utilise land available with us instead of purchasing separate. Here the location chosen is Kothamangalam. For building, we are going with lightweight PEB (Pre-Engineered Building) shed with steel trusses and GI roofing.

Total cost per product

Electronics	3680
Packaging	400
Variable cost	5%
Total	**₹4284**

Initial investment

Building cost	**4,65,000**
Equipments	67,000
Clearance charges	~52,000
Initial Material Cost	85680
Buffer	20,000
Total	**₹6,89,680**

per-month operational cost

Electricity	8,000
Water	1,000
Insurance	4,000
Maintenance	1,000
Total	**₹14,000**

We are planning to pool initial investment from us founders. Each one is investing 1.38L. One of us is providing land. Hence the partnership contribution adjusts accordingly.

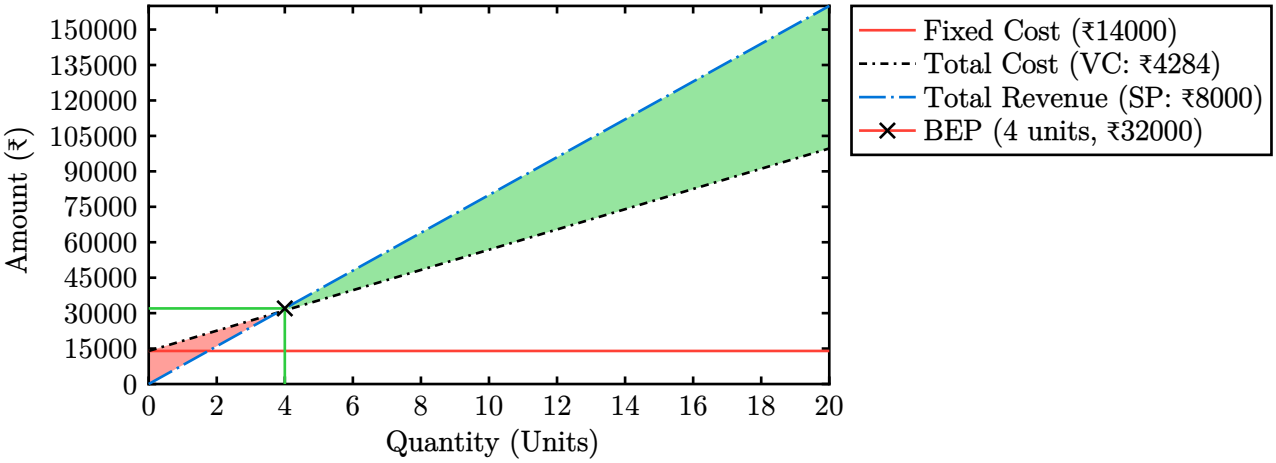
We are setting a Selling Price of: ₹8000/unit.

Here contribution margin is ₹8000 − ₹4284 = ₹3716

Break Even Point (BEP)

$$\text{BEP in units} = \frac{14000}{3716} \approx 4 \text{ units/month}$$

$$\text{BEP in revenue} = 8000 * 4 = ₹32000$$



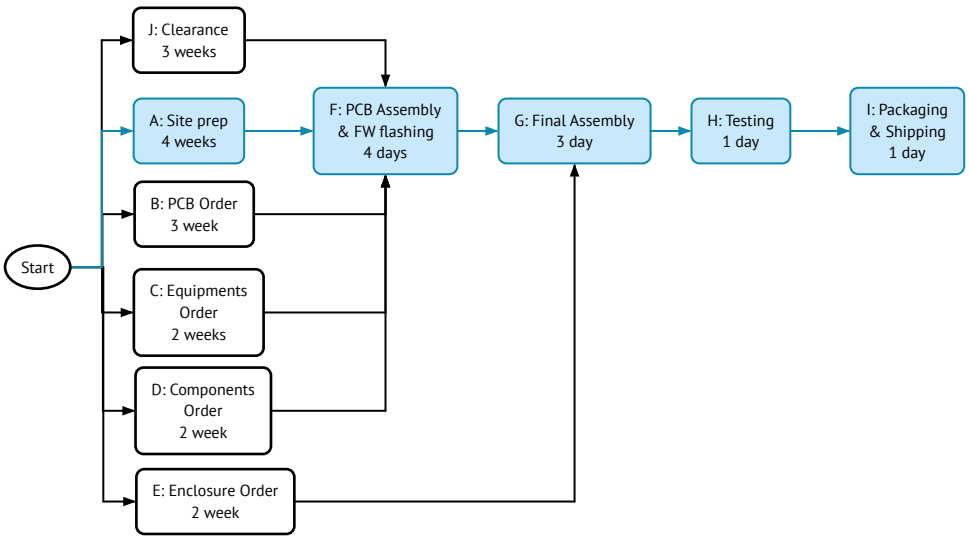
We are expecting sales of 20 units/month; in this case the initial investment is expected to be recovered by :

$$\frac{₹689680}{₹3716/\text{unit} * (20 \text{ units} - 4 \text{ units})} \approx 12 \text{ months}$$

After the payback the net profit margin is:

$$\frac{\text{Net income}}{\text{Net revenue}} = \frac{₹3716 \times 20 - ₹14000}{₹8000 \times 20} \times 100\% = \mathbf{37.7}\%$$

CPM diagram



Critical Path is: $A \to F \to G \to H \to I$

Total estimated project Duration is 37 Days.